Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Quantity

Meaning in my words	How can you tell?
What do physicists mean by a quantity?	Two labels say 4 m and 4 m/s. What different questions could they answer?
Example or sketch	**Non-example and why**
What label would make "−3" into a useful velocity reading?	A student writes "force = 6 m." What clue tells you to check their work?

Sign

Meaning in my words	How can you tell?
How would you explain sign to a friend?	If right is positive, what sign goes on a 3 N force left?
Example or sketch	**Non-example and why**
How would you label a leftward velocity of 2 m/s when right is positive?	Does a negative velocity by itself mean the cart is slowing down?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force

Meaning in my words How would you explain force to a friend?	**How can you tell?** When your hand pushes a cart, what pushes and what receives the push?
Example or sketch Which force arrow would you draw while a hand pushes a cart right?	**Non-example and why** The hand lets go and the cart keeps rolling. Is the hand still pushing it?

Check

Meaning in my words How would you explain check to a friend?	**How can you tell?** You found displacement using an equation. What feature of a velocity–time graph could check it?
Example or sketch A cart stays at +3 m/s for 2 s. How can a graph check the result of 6 m?	**Non-example and why** Someone copies the same number twice. What could actually catch a wrong answer?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

First wrong step

Repair

My marks mean

Stop the faulty graphic

19

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use all forces to calculate acceleration.

CHOOSE ONE JOB TOGETHER

SOFTWARE TESTER ☐

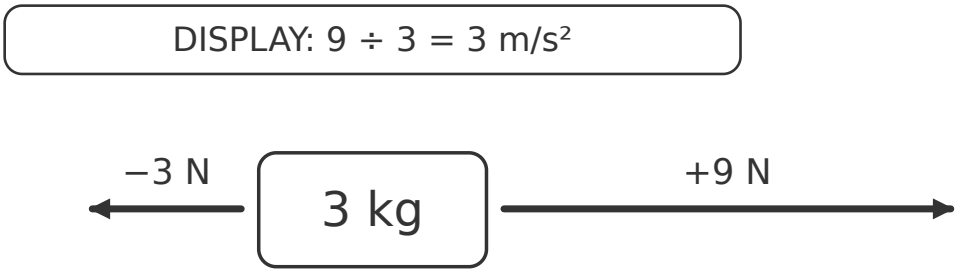
Mark the first wrong calculator step. Send the developer the corrected force sum and expected acceleration.

FACT-CHECK PRODUCER ☐

Stop or clear the graphic. Give the anchor a corrected number that viewers can check against the force arrows.

Point to a clue. Keep your first marks; add the update using the color key.

A calculator overlay for a **3 kg** tracker shows **+9 N** and **–3 N**. Its output is "**9 ÷ 3 = 3 m/s².**" It is about to go live.



01 OUR FIRST CALL

Can this calculation go live? Give the corrected result and explain.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Did the developer fix the error? Send a follow-up that names what still needs changing.

Switch speakers. Create a different pair of opposing forces that would expose the same calculator error. Show the correct result.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks